ATEM MCP Server

Quick manual: control a Blackmagic ATEM switcher by talking to Claude | Set up for the Constellation 4K, with EDL logging for DaVinci Resolve

1. Setup

Needs: a Mac on the same network as the ATEM, Node.js 22 or newer, and Claude Desktop (best for live shows) or Claude Code. The server is already installed and built in `~/mcp-servers/atem-mcp-server`.

Step	What to do
1. Find the ATEM's IP	Open **ATEM Setup**, select the switcher, and check the Configure tab (for example 192.168.1.240). The Mac must be on that same network or subnet.
2. Add it to Claude Desktop	Quit Claude Desktop fully (**Cmd+Q**), then run in Terminal: `~/mcp-servers/atem-mcp-server/add-to-claude-desktop.sh 192.168.1.240` Use your ATEM's IP. Reopen Claude Desktop. It backs up your config first.

3. Claude Code (optional)	Already registered. To add auto-connect, run:
	`claude mcp remove atem -s user`
	`claude mcp add atem -s user -e ATEM_HOST=192.168.1.240 -- node ~/mcp-servers/atem-mcp-server/dist/index.js`

4. Test it	In a new chat: *"Show me the switcher status"*. You should see the model name, the inputs, and what's on program and preview.

Fresh install on another Mac:

```
git clone https://github.com/guycochran/atem-mcp-server.git ~/mcp-servers/atem-mcp-server
cd ~/mcp-servers/atem-mcp-server
npm install --include=dev && npm run build
(the EDL logger is a local addition: copy src/services/edl-logger.ts and src/tools/edl.ts
across, then rebuild)
```

2. What you can do, and what to say

Say these to Claude in plain English. You can combine them in one request (*"preview camera 2 and dissolve to it"*). Input numbers are the ATEM input numbers: camera 1 = input 1.

Feature	Example commands	Notes
Connect / status	*"Connect to my ATEM at 192.168.1.240"* *"Show switcher status"*	Connects automatically if ATEM_HOST is set
Switching	*"Cut to camera 3"* *"Put camera 2 on preview"* *"Preview camera 4 and dissolve to it"* *"Fade to black"*	M/E 1 by default
Transitions	*"Set transition to mix, 30 frames"* *"Use a wipe"* *"Move the T-bar to 50%"*	Mix, dip, wipe, DVE, stinger
Aux routing	*"Route camera 1 to aux 1"* *"What is on aux 2?"*	Confidence monitors, record feeds
Keyers	*"Put DSK 1 on air"* *"Auto DSK 1"* *"Set DSK 1 fill to media player 1"* *"Upstream key 1 on"*	Lower thirds, bugs, PiP

Feature	Example commands	Notes
Macros	*"List macros"* *"Run macro 3"* *"Stop the macro"*	Runs macros already on the switcher
Super Source	*"Side-by-side with cameras 1 and 2"* *"Go gallery with guests on 2, 3 and 5"* *"2x2 grid"*	Only if your model has Super Source
Saved looks	*"Save this look as podcast-2guest"* *"Load podcast-2guest with cameras 1 and 4"* *"List my looks"*	Saved on the Mac in ~/.atem-mcp/looks
Auto-switching	*"Cut to active speaker"* *"Auto switch"* *"Auto switch host + active speaker"* *"Auto switch off"*	Follows whoever is talking, using audio levels
Audio mixer	*"Lower camera 2 audio by 5 dB"* *"Mute input 3"* *"Camera 1 audio follow video"* *"Master to -3 dB"*	
Fairlight EQ / dynamics	*"Podcast EQ on mics 1 and 2"* *"Compress mic 1 4:1 at -20"* *"Gate mic 2 below -40"* *"Limiter on master at -3"* *"Reset EQ on input 1"*	Presets: vocal, podcast, music, de_mud, flat
Record / stream	*"Start recording"* *"Stop streaming"* *"Recording status?"*	Only on models with a built-in recorder or streamer (Mini Pro and up). **Not on Constellation 4K.**
EDL logging	*"Start the EDL for Episode 12"* *"Mark that, clip for social"* *"Is the EDL running?"* *"Stop the EDL"*	See section 3

Special input numbers

Input ID	Source	Input ID	Source
1-20	Physical inputs	3010 / 3020	Media player 1 / 2
1000	Color bars	6000	Super Source
2001 / 2002	Color generator 1 / 2	10010	Black

3. EDL logging for DaVinci Resolve

Logs every program cut on M/E 1, whether it comes from the panel, ATEM Software Control, Claude or auto-switch. It writes a CMX 3600 .edl, and the file updates after every cut, so a crash loses nothing. Files are saved to `~/Movies/ATEM EDLs/`.

When	Do this
Before the show	**Sync timecode.** Set the ATEM and every camera or recorder to the same time-of-day timecode (jam-sync if possible). If the recorders follow the Mac's clock, say *"start the EDL using the Mac clock"*. **Name the inputs.** Reel names come from the ATEM input short names (CAM1, CAM2...). Rename them in ATEM Software Control, or say *"reel names: input 1 is A001, input 2 is B001"*.

When	Do this
During the show	Roll the cameras first, then *"Start the EDL for [show name]"*. Add markers with *"Mark that..."*. Stop the EDL before the cameras stop.
In Resolve	1. Import the camera files, and set each clip's **Reel Name** to match the EDL (Clip Attributes, or Project Settings, General Options, 'Assist using reel names'). 2. **File, Import, Timeline**, then choose the .edl. Point it at the camera clips and match by reel name. 3. Markers: right-click the timeline, choose **Timelines, Import, Timeline Markers from EDL**, then pick the _markers.edl file.
Limits	Logs the program bus only: keyers, DSK graphics and Super Source composites aren't separate layers. All transitions are written as dissolves of the measured length.

4. Live show tips

Tip	
Use Claude Desktop	Claude Code asks for approval on every command, which is too slow for live switching.
Pre-show check	*"Show switcher status and the audio mixer state"*. Then test a cut and a dissolve, and set your transition style and rate.
Be specific	*"Cut to camera 3"* works faster than a long sentence. Expect about a second of delay.
Keep a fallback	Keep the hardware panel or ATEM Software Control open. Claude is best for setup, looks, audio and auto-switching.
Remote control	Phone or claude.ai control needs a Cloudflare tunnel. The server's built-in login lets in anyone who has the URL, so don't expose it without adding real auth.

5. Troubleshooting

Problem	Fix
No ATEM tools in Claude Desktop	Quit fully with Cmd+Q and reopen. Check the logs in ~/Library/Logs/Claude/mcp*.log, and re-run the add-to-claude-desktop.sh script.
Can't connect to the ATEM	Is the Mac on the same network? Run `ping 192.168.1.240`. The port is UDP 9910. Check the IP in ATEM Setup.
A command does nothing	Some features depend on the model. Ask *"show switcher status"* to check the connection and what the switcher has.
EDL won't relink in Resolve	The camera timecode doesn't match the switcher's, or the reel names differ. Check one clip's start timecode against the EDL's source timecode.
Server build fails (tsc not found)	Run `npm install --include=dev`, then `npm run build`.